Distribution metrics between a protein and its mutants, decomposed by site and by mode

Notes for penm

2026-10-16

1 What is shown

For every single-point mutant of a complete genm scan, several metrics between the wild type's and the mutant's Boltzmann distributions are computed and split into per-site and per-mode contributions. The figures are meant to give a feeling for how the metrics behave; they recommend none.

Data. 2acy, the genm vignette's wild type (`node = "ca"`, `model = "ming_wall"`, $d_{\max} = 10.5$ Å, `d_max_graph = 14`, `ensemble = 1`, `alphabet_size = 10`, `mut_dl_sigma = 0.3`, `mut_sd_min = 2`); all 882 single-point mutants (98 sites, 9 letters each); β at 298 K. One protein, one model, one ensemble. Scripts: `metrics_scan.R` (computation) and `metrics_plots.R` (figures and numbers); their printed output is in `metrics_scan.out.txt` and `metrics_plots.out.txt`.

2 Definitions

Both distributions are Gaussians on the wild type's internal subspace; genm superposes the mutant onto the wild type. In the wild type's mode coordinates ($\mathbf{U}$, $3N \times d$, $d = 3N - 6 = 288$):

$$\mathbf{A} = \mathrm{diag}(\sigma_n^2),\ \sigma_n^2 = \frac{1}{\beta\lambda_n}; \qquad \mathbf{B} = \mathbf{U}^T\mathbf{C}^{\mathrm{mut}}\mathbf{U}; \qquad \delta\mathbf{r} = \mathbf{U}^T(\mathbf{r}^{\mathrm{mut}} - \mathbf{r}^{\mathrm{wt}}).$$

A contribution matrix $\mathbf{M}$ in mode coordinates is attributed to mode n by M_{nn} and to site i by the trace of the 3×3 block of site i in $\mathbf{U}\mathbf{M}\mathbf{U}^T$. A vector $\mathbf{z}$ is attributed by z_n^2 and $|(\mathbf{U}\mathbf{z})_i|^2$. Both attributions sum to the same total, which the scan checks for every mutant.

Wasserstein-2 (Å^2). $W_2^2 = |\delta\mathbf{r}|^2 + \mathrm{tr}\big(\mathbf{A} + \mathbf{B} - 2(\mathbf{A}^{1/2}\mathbf{B}\mathbf{A}^{1/2})^{1/2}\big)$. It is $E|\mathbf{x} - T(\mathbf{x})|^2$ under the optimal coupling, with $\mathbf{T} = \mathbf{A}^{-1/2}(\mathbf{A}^{1/2}\mathbf{B}\mathbf{A}^{1/2})^{1/2}\mathbf{A}^{-1/2}$. Mean part: $\delta\mathbf{r}$, attributed as a vector, so by site it is $|\delta\mathbf{r}_i|^2$ (penm's `dr2i`) and by mode δr_n^2 (`dr2n`). Covariance part: $\mathbf{M} = (\mathbf{I} - \mathbf{T})\mathbf{A}(\mathbf{I} - \mathbf{T})^T$.

Spectral divergences (nats). With the whitened matrix $\mathbf{X} = \mathbf{A}^{-1/2}\mathbf{B}\mathbf{A}^{-1/2} = \mathbf{Y}\,\mathrm{diag}(\nu_k)\mathbf{Y}^T$, the covariance part of each is $\sum_k g(\nu_k)$, attributed through $\mathbf{M} = \mathbf{Y}\,\mathrm{diag}(g(\nu_k))\mathbf{Y}^T$:

divergence	$g(\nu)$	near $\nu = 1 + x$
KL(mut‖wt)	$\frac{1}{2}(\nu - 1 - \ln \nu)$	$x^2/4$
Jeffreys (symmetrised KL)	$\frac{1}{2}(\nu + 1/\nu - 2)$	$x^2/2$
Bhattacharyya	$\frac{1}{2}\ln\big((1+\nu)/(2\sqrt{\nu})\big)$	$x^2/16$
affine-invariant (Riemannian), squared	$\ln^2 \nu$	x^2

Only the KL mean part is shown: $\frac{1}{2}|\mathbf{A}^{-1/2}\delta\mathbf{r}|^2 = \frac{\beta}{2}\,\delta\mathbf{r}^T\mathbf{K}\delta\mathbf{r}$, attributed as the vector $\mathbf{A}^{-1/2}\delta\mathbf{r}$; by site this is exactly $\frac{\beta}{2}$ `de2i` (checked on one mutant), by mode $\frac{\beta}{2}$ `de2n`. Site attribution uses the symmetric square root $\mathbf{C}_{\mathrm{wt}}^{-1/2}$, so site shares are in units of the wild type's own fluctuations.

Frobenius (Å^4, covariance part only). $\|\mathbf{B}-\mathbf{A}\|_F^2$; by mode $\sum_m (B-A)_{nm}^2$; by site $\sum_j \|\Delta\mathbf{C}_{ij}\|_F^2$, the row of 3×3 blocks of the Cartesian difference, so correlations with other sites count.

Sums of marginal divergences. KL and W_2^2 between the 3×3 marginals of each site, and between the one-dimensional marginals along each wild-type mode, means included. These add up by construction, but their sum is not the divergence between the full distributions.

penm. `dmsfi`^{2} and `dmsfn`^{2}, squared as in the genm vignette.

3 Results

3.1 Averages over the scan, by site and by mode

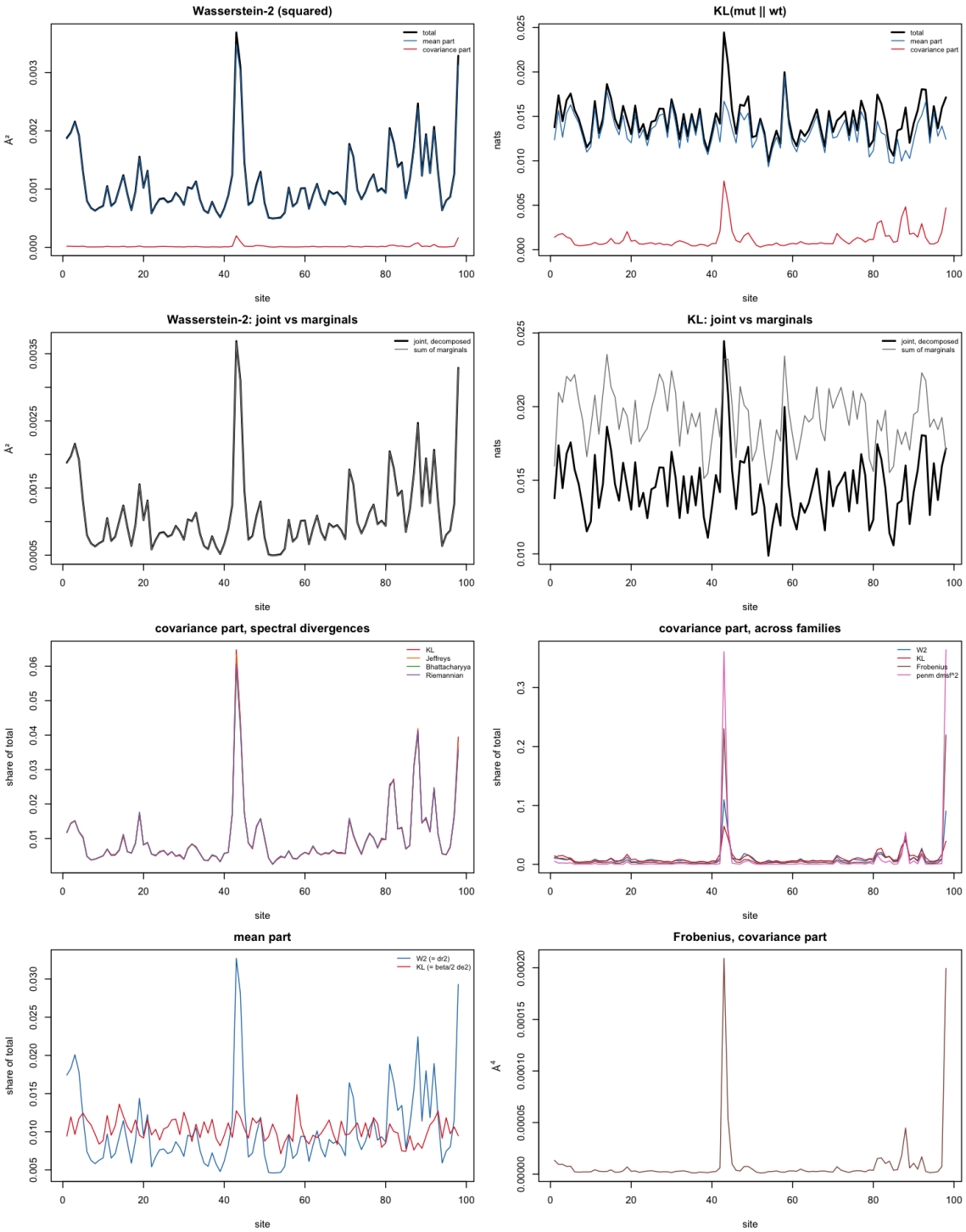


Figure 1: Means over the 882 mutants, by site. "Share of total" panels divide each profile by its sum, to compare shapes across units.

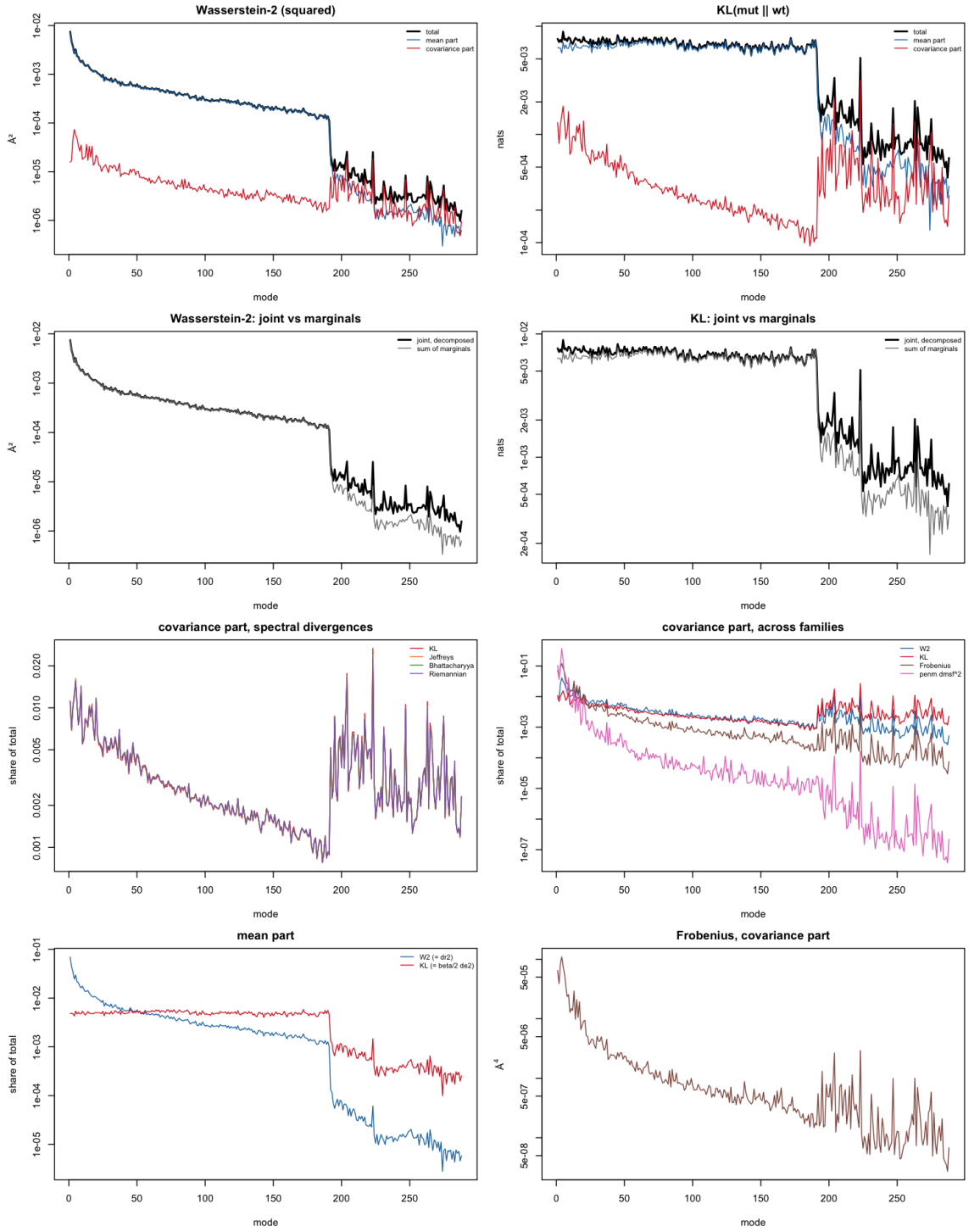


Figure 2: As Figure 1, by wild-type mode; logarithmic vertical axes.

What the two figures show:

- **Mean parts dominate.** Median over mutants of the covariance part's share of the total: 1.0% for W_2^2, 7.2% for KL. Across mutants the KL mean and covariance parts correlate

(Spearman 0.58).

- W_2 **weighs by displacement, KL by energy.** The W_2 mean part by site is `dr2i`: largest at the flexible sites (around 43–44, the C terminus, 80–98). The KL mean part by site, $\frac{\beta}{2}$`de2i`, is nearly flat. The two site profiles have Spearman 0.22. By mode, W_2 falls with mode index (soft modes move most), while the KL mean part is nearly flat across the lower band: there $\frac{1}{2}\delta r_n^2/\sigma_n^2$ is about constant, so the mutational displacements along the lower-band modes scale like the thermal variances σ_n^2. Both drop at the band gap (mode 191/192).
- **The four spectral divergences give the same profiles.** Their scan-mean profiles have Spearman 1.00 with each other, by site and by mode. Near $\nu = 1$ all four g are proportional to $(\nu - 1)^2$; over all mutants ν ranges over 0.24–4.4, but the median over mutants of the largest $|\nu - 1|$ is 0.32.
- **Covariance parts across families.** By site, the covariance parts of W_2, KL, Frobenius and `dmsf`^{2} peak at the same sites (43 and 98 above all), with Spearman 0.86–0.91, but differ in how concentrated they are: the shares of sites 43 and 98 are 0.36 each for `dmsf`^{2}, 0.23 and 0.22 for Frobenius, 0.11 and 0.09 for W_2, while KL's largest shares are 0.065 (site 43) and 0.044 (site 44). By mode they differ more: W_2, Frobenius and `dmsf`^{2} fall with mode index, `dmsf`^{2} by about five orders of magnitude, while KL is flatter and rises in the upper band (Spearman of KL with `dmsf`^{2} by mode: 0.42).
- **Joint against marginals.** For W_2, the per-site attribution of the joint and the sum of site marginals nearly coincide: the mean part is the same in both, $|\delta\mathbf{r}_i|^2$, and the covariance part is small. For KL, the sum of site marginals exceeds the joint (median ratio 1.27) and its site profile follows the KL mean part (Spearman 0.87) but not the joint covariance part (0.13): each marginal weights $\delta\mathbf{r}_i$ by its own block $\mathbf{C}_{ii}^{-1}$, not by the joint's $\mathbf{K}$.

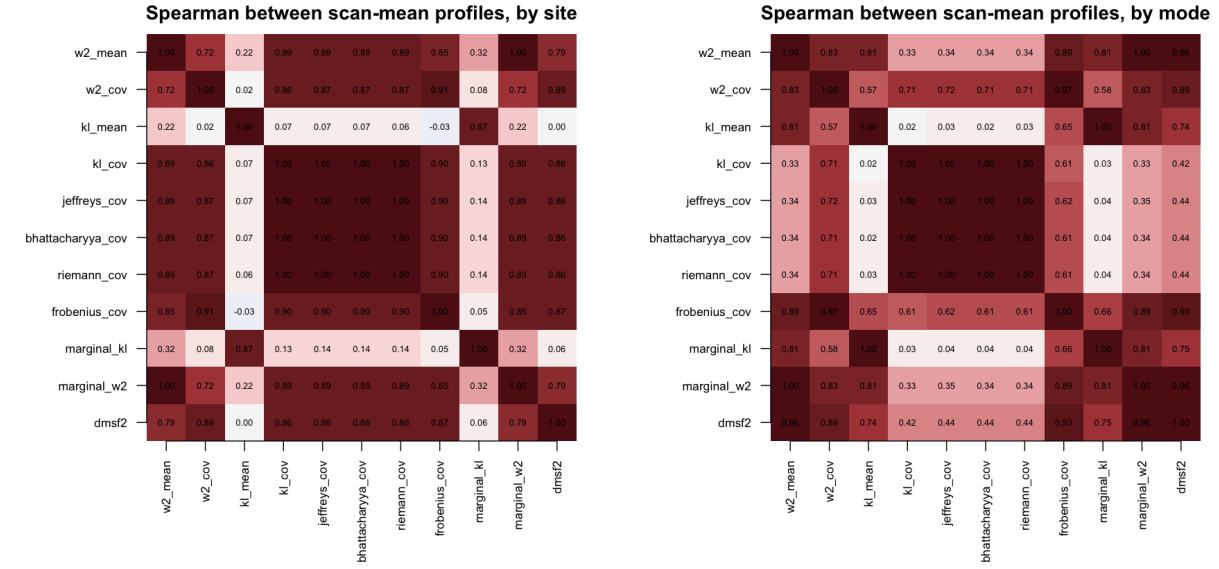


Figure 3: Spearman correlations between the scan-mean profiles of all components, by site (left) and by mode (right).

3.2 Individual mutants

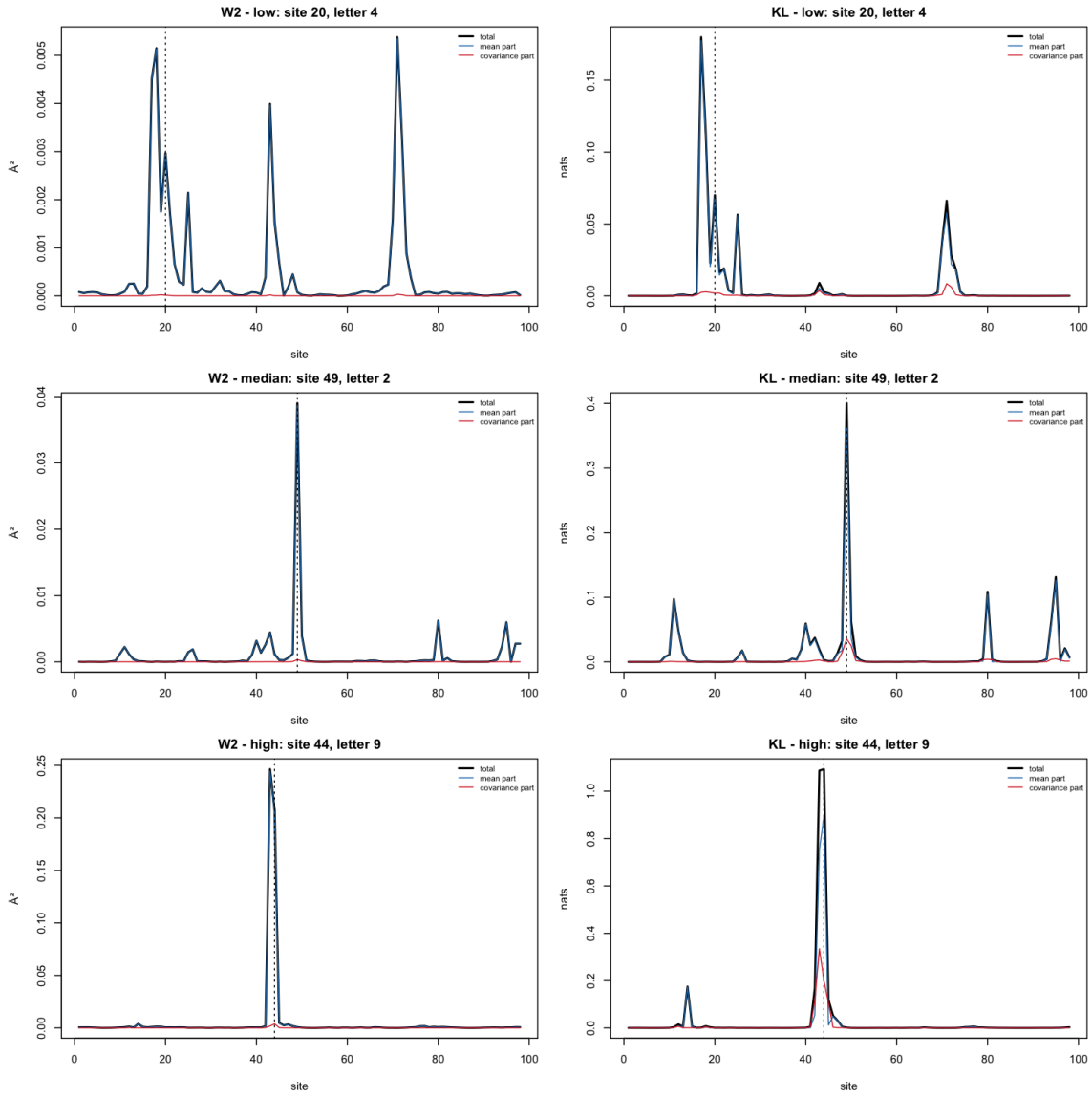


Figure 4: Site profiles of W_2^2 and KL for three mutants: the 10th percentile (site 20, letter 4), the median (site 49, letter 2) and the largest (site 44, letter 9) of W_2^2. The dotted line marks the mutated site.

Single mutants have localised profiles: the mutated site and a few others. W_2 and KL agree on where the peaks are but not on their heights; in the 10th-percentile mutant, the peaks near sites 18 and 71 are about equal in W_2, while in KL the one near 17–18 is nearly three times the one at 71. The covariance part is visible only in KL, and mostly next to the mutated site (in the largest mutant, about 0.33 nats at site 43).

3.3 Totals over mutants

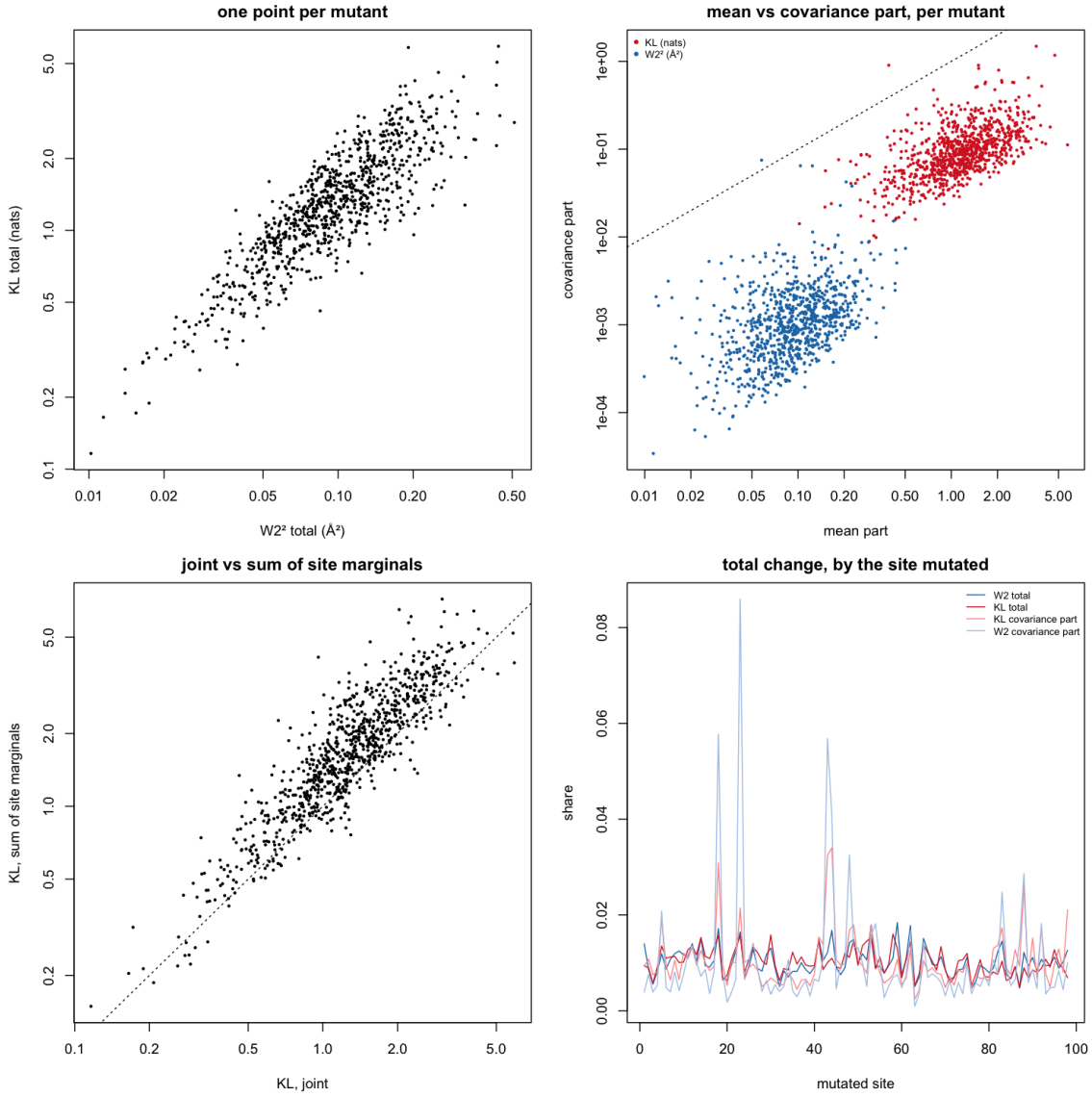


Figure 5: Top left: KL against W_2^2, one point per mutant. Top right: covariance part against mean part, for both. Bottom left: KL between the full distributions against the sum of site-marginal KLs; the dotted line is equality. Bottom right: each total averaged over the nine mutations at each site, as a share of its sum.

- Global totals: W_2^2 median 0.093 Å^2 (mean part) and 0.0011 Å^2 (covariance part); KL median 1.17 and 0.091 nats.
- The two totals rank mutants similarly (Spearman 0.84).
- Averaged by the mutated site, the totals are fairly even across sites; the covariance parts are not: in the figure, W_2's covariance part is largest for mutations around sites 18, 23, 43–44, 48, 83 and 88.

4 Things to keep in mind

- Under lfenm the covariance is unchanged, so every covariance part is zero and each metric reduces to its mean part.
- The KL and spectral attributions to sites depend on choosing the symmetric square root for whitening; W_2's and Frobenius's do not involve a choice.
- Every comparison is made in the wild type's internal subspace, after genm's superposition.
- One protein, one elastic network model, one ensemble of mutations.